Quantum Reservoir Computing for Zero-Day Network Intrusion Detection

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1. Problem statement

New attack types appear every day and AI is accelerating them: 16% of breaches now involve AI-driven attackers, the average attacker breakout time is 29 minutes, and the average breach costs US$4.88M. Signature-based and supervised defences must see labelled examples of an attack before they can catch it, so a zero-day attack is invisible to them, and retraining a deep network for every new attack family is slow and costly. Security teams need a detector that **(i) flags behaviour it has never seen, (ii) learns from normal traffic alone, with little data, and (iii) is cheap to retrain.**

2. Approach

- **Quantum reservoir.** A fixed, randomly drawn trapped-ion circuit (transverse-field Ising dynamics with the all-to-all ZZ couplings that ion traps provide natively) acts as a nonlinear feature map. Only a linear readout is trained, so the model is updated in seconds with no gradients through the quantum circuit.
- **Encoding and memory.** Each network flow's features become rotation angles on half of the qubits over L time steps. These input qubits are measured and reset mid-circuit at every step, while the memory qubits are never measured and carry information forward. Averages over 512 shots form the feature vector.
- **Two modes.** Novelty mode: a distance detector fitted on benign traffic only, for zero-day attacks. Classify mode: a supervised linear classifier for known attack families.
- **Implementation.** Circuits use Quantinuum's native operations (PhasedX, ZZPhase, mid-circuit measure-and-reset), are simulated exactly (state vector with sampled shots) for the full study, executed on the H2-Emulator through Aqora, and costed per flow in H2 HQC.
- **Benchmark design.** NF-UNSW-NB15-v2: 241,165 unique flows, 53 inputs, 17.2% attacks. 18 reservoir sizes, 6–16 qubits × L = 6, 8, 10 steps. 400 training and 300 test flows per configuration; the Exploits family is withheld from training entirely and used as the zero-day test (150 flows). Baseline: the identical readout on the classical features with no quantum layer. Every quantum-minus-classical difference carries a paired 95% bootstrap interval (4,000 resamples).

3. Key results

Qubits	2-qubit gates per flow	Est. error-free run on H2	Zero-day AUC gap	Known-attack AUC gap
6–8	105–308	91–77%	−0.017	−0.009
10–12	315–726	77–55%	−0.010	−0.007
14–16	637–1,320	59–34%	−0.003 (95% CI −0.012 to +0.007)	−0.012

Gap = reservoir AUC minus classical AUC, averaged over the configurations in each row (classical: zero-day 0.888, known attacks 0.986). Error-free estimate = product of H2 gate fidelities.

- **No configuration beats the classical baseline.** We report this as measured: on known attacks the reservoir is behind in all 18 configurations, by 0.001–0.024 AUC.
- **On zero-day attacks the reservoir catches up as it grows:** +0.0017 AUC per added qubit (95% CI +0.0005 to +0.0031), and L = 10 beats L = 8 by +0.008. At 14–16 qubits it is level with the baseline (best: 16 qubits, L = 8, 0.891 vs 0.888), not ahead of it.
- **Much of the gain is input capacity.** The 3 configurations with fewer encoding slots than the 53 inputs average −0.025 on zero-day; the 15 that read every input average −0.007 (difference +0.018, 95% CI +0.004 to +0.032).
- **Data representation mattered most.** Moving to the richer v2 features lifted the classical zero-day AUC from 0.827 to 0.888 and the reservoir's from 0.761–0.789 to 0.859–0.882.
- **Validation.** A full re-run reproduced all 12 repeated configurations bit-for-bit, and parallel runs match single-core runs exactly. Detection at 1% false alarms is not claimed: with 150 benign test flows its interval is about ±16 percentage points.

4. Relevance to quantum computing

The quantum-relevant finding is the size trend: the zero-day gap closes as the register grows and reaches parity at 14–16 qubits, where classical simulation is already expensive (one 20-qubit configuration needs about 50 hours on our machine). Whether the reservoir moves past parity can only be tested above 20 qubits, which is the regime of Quantinuum hardware. The same data price that step honestly: the best-scoring circuits are the deepest, with a 34–59% estimated chance of an error-free run and 700–1,450 HQC per flow at 512 shots, so noise and shot budget, not qubit count, are the binding constraints. **Executed on Quantinuum's stack:** 3 reservoir circuits (6 qubits, L = 6, 105 ZZPhase gates and 18 mid-circuit measure-and-resets each, 200 shots) ran on the H2-Emulator; its measured features match our simulator with r = 0.994, all 72 within 3σ of shot noise. **Phase 2:** scale these emulator runs to the 8–16 qubit circuits and full test sets, repeat over several data splits and reservoir draws, and enlarge the benign test set so detection at 1% false alarms can be measured.

5. Potential impact

- **Market.** Network detection and response: US$4.04B global (TAM), US$315M APAC (SAM), US$25.0M Singapore (SOM), led by government and defence (38%), banking (24%) and telecom (22%).
- **Deployment.** Novelty mode runs as an early-warning layer beside existing detection tools. It is trained on an organisation's own benign traffic, needs no labelled attacks, retrains in seconds when the network changes, and every analyst verdict feeds back as a new training row.
- **Value.** Catching an unseen attack inside the 29-minute breakout window protects against an average US$4.88M breach, which a signature-based defence cannot prevent for an attack it has never seen.

Sources: industry and market figures as cited in our presentation (slides 2 and 8). Full methods, per-configuration results and error analysis: QRC_v2_report_full_grid_6to16_qubits.pdf. Code, results and README: github.com/BinhDangNguyn/iontraps_qrc. H2-Emulator job IDs: AFByb3ZpZGVySm9iAaE7UeXMfVCXVYSUU-KRFA, AFByb3ZpZGVySm9iAaE7UerBfkKVNjaa_Qh69Q, AFByb3ZpZGVySm9iAaE7Ue9Ze1CP_cVW5SsOcA.